

UNIVERSITY OF TORONTO MISSISSAUGA
APRIL 2025 FINAL EXAMINATION
MAT102H5S - Introduction to Proofs
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Duration - 3 hours
Aids: None

Name: _____
Student Number: _____
Signature: _____

This exam contains 12 pages (including this cover page) and 7 problems. Each problem is worth 10 marks, making the exam worth 70 marks. Check to see if any pages are missing and ensure that all required information at the top of this page has been filled in. You must also have the corresponding Multiple Choice Booklet, which contains the multiple choice questions needed for Question 7 (page 12).

Each part of Question 7 includes five multiple choice answers, of which only a single response is correct. All answers should be indicated on the included scantron sheet (page 12). When filling out the scantron:

- Use a #2 pencil to bubble answers (do not use ink). A #2 pencil ensures that the marks are dark enough.
- Ensure that bubbles are completely filled and that erased answers are completely erased.

Any failure to follow these instructions which results in the scanner not being able to read your work will result in a deduction of 3 marks. You will not be penalized for incorrect answers.

Your solutions to Problems 1 - 6 must be written in the space provided. There are two additional pages (pages 10 and 11) at the end of the booklet for any overflow work. Clearly indicate in your solution if your work will continue onto that page. No additional paper may be submitted for grading. Do not, under any circumstances, write on the QR code in the top right corner of each page. **Doing so will result in you receiving a zero (0) on this exam.**

The University of Toronto Mississauga and you, as a student, share a commitment to academic integrity. You are reminded that you may be charged with an academic offence for possessing any unauthorized aids during the writing of an exam. Clear, sealable, plastic bags have been provided for all electronic devices with storage, including but not limited to: cell phones, SMART devices, tablets, laptops, calculators, and MP3 players. Please turn off all devices, seal them in the bag provided, and place the bag under your desk for the duration of the examination. You will not be able to touch the bag or its contents until the exam is over.

If, during an exam, any of these items are found on your person or in the area of your desk other than in the clear, sealable, plastic bag, you may be charged with an academic offence. A typical penalty for an academic offence may cause you to fail the course.

Please note, once this exam has begun, you **CANNOT** re-write it.

1. Define the sets

$$A = \{f : \mathbb{N} \rightarrow \mathbb{N} : \exists N \in \mathbb{N}, \forall n \geq N, f(n) = 0\} \quad \text{and} \quad B = \{f : \mathbb{N} \rightarrow \mathbb{N} : \forall N \in \mathbb{N}, \exists n \geq N, f(n) = 0\}.$$

(i) (3 points) Show that $A \subseteq B$.

(ii) (3 points) Show that the inclusion in part (i) is strict; namely, that $A \neq B$.

- (iii) (2 points) If $f, g : \mathbb{N} \rightarrow \mathbb{N}$ are functions, we define their sum $f+g : \mathbb{N} \rightarrow \mathbb{N}$ as $(f+g)(n) = f(n)+g(n)$ for all $n \in \mathbb{N}$. Show that A is closed under addition; that is, if $f, g \in A$ then $f+g \in A$. Be sure to justify your answer.

- (iv) (2 points) With addition defined as in part (iii), is B closed under addition? Be sure to justify your answer.

2. Consider the following set $S \subseteq \mathbb{Z}$, which we define recursively:

- $0 \in S$
- If $n \in S$ then $2n + 1 \in S$.

(i) (3 points) Write out six (6) elements of this set.

(ii) (7 points) Prove that $s \in S$ if and only if $s = 2^n - 1$ for some non-negative integer n .

3. For this problem, p is a prime number and we are working in the field $\mathbb{Z}_p$.

- (i) (5 points) If $[a] \in \mathbb{Z}_p$, show that $[a] = [a]^{-1}$ if and only if $[a] = [1]$ or $[a] = [p - 1]$. *Hint:* Note that $[a] = [a]^{-1}$ if and only if $a^2 \equiv 1 \pmod{p}$.

- (ii) (5 points) Show that $[(p - 1)!] = [p - 1]$. *Hint:* If $[a]$ and $[b]$ are multiplicative inverses, then $[a][b] = [1]$. Use the fact that $\mathbb{Z}_p$ is a field, together with part (i), to conclude the result.

4. Suppose that $A \subseteq \mathbb{N}$ is a non-empty set, and define a relation $\sim$ on A as

$$a \sim b \quad \text{if and only if} \quad a \mid b \text{ or } b \mid a.$$

(i) (2 points) Show that for any set A , $\sim$ is symmetric and reflexive.

(ii) (4 points) Determine whether $\sim$ is an equivalence relation on $A = \mathbb{N}$.

(iii) (4 points) Determine whether $\sim$ is an equivalence relation on $A = \{2^n : n \in \mathbb{N}\}$.

5. (10 points) Suppose that S is a set. Given a function $f : S \rightarrow S$ and a positive integer n , we define

$$f^{\circ n} = \underbrace{f \circ f \circ \cdots \circ f}_{n\text{-times}}.$$

Suppose that $f : S \rightarrow S$ satisfies $f^{\circ n} = f$ for some integer $n \geq 2$. Show that f is injective if and only if it is bijective. Is the same statement true if we replace “injective” with “surjective”?

6. (i) (3 points) Recall that for any sets S and T , we define $|S| + |T| = |S \times \{0\} \cup T \times \{1\}|$. Show that $|S| + |T| \geq |S \cup T|$.

- (ii) (3 points) Suppose that S and T are *finite* sets such that $|S| + |T| > |S \cup T|$. Show that $S \cap T \neq \emptyset$.

- (iii) (4 points) Suppose that $n \geq 2$ is a positive integer and A is a subset of $\{1, 2, \dots, n-1\}$ such that $|A| \geq n/2$. Show that there exist $a_1, a_2 \in A$ such that $a_1 + a_2 = n$. *Hint:* Use part (ii), where one of your sets is $T = \{n - a : a \in A\}$.

This page is for additional work and will not be marked unless indicated.

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End of Exam Questions.